

Fig. 3: Block Diagram after clean up

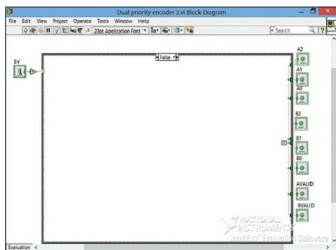


Fig. 4: All Tunnels closed

used in this project; this will be considered as main Case Structure.) The labels in the menu at the top of the Case Structure will appear as True and False. First consider the True condition.

Step 9. Select Programming-> Boolean-> NOT gate and drop it on to Block Diagram. Place the cursor at output of vertical toggle switch (En). You will see a hot spot. From there pull a wire and give it as input to NOT gate. The output of NOT gate is connected to the selector terminal '?' of the Case Structure.

Step 10. Again, select Programming-> Boolean-> Bool to (0, 1) function and drop it on to Block Diagram. Connect the output of I7 to input of Bool to (0, 1) function.

Step 11. Select Programming-> Comparison-> Equal ? function and

drop it on to Block Diagram. Give the output of 'Bool to (0, 1)' to one of the inputs of 'Equal ?'. Place cursor on the second input of 'Equal ?' and right click on it. Select Create->Constant and put value 1.

Step 12. Select Programming-> Comparison-> Select function and drop it on to Block Diagram. It has three input lines, namely, true (t), select (s) and false (f). Connect the output of 'Equal ?' to 's' input of Select function.

Step 13. Repeat steps 10, 11 and 12 for remaining inputs (I6, I5...I1), except for I0.

Step 14. Place cursor on the true (t) input of Select function, right click on it and create a constant of value 7 for input I7. Similarly, do it for all inputs with constant value decreased by 1 for each input of Select function, which are arranged in descending order as shown in Fig. 2.

Step 15. Connect the output of Select function corresponding to I1 to false (f) input of Select function corresponding to I2. Similarly, connect the remaining outputs of Select function to false (f) input of the Select function, except for I7. Create a constant 0 for the false (f) input of Select function of input I1.

Step 16. Add two Case Structures as explained below. Select Programming-> Structures-> Case Structure, drag and drop it on to Block Diagram

inside the main Case Structure. If labels in the menu appear as True and False, modify them to 0 and 1, respectively. Right click on case selector or menu, select 'Add Case After' option and label it as 2. Similarly, do the same for the other five cases and label them as 3, 4, 5, 6 and 7.

Step 17. Connect the output of the Select function of I7 to selector terminal '?' of the inner upper Case Structure.

Step 18. Select case 0 and then select Programming->Boolean-> False Constant and connect the output of False Constant (labeled as F) to the Round LED A2, which is placed outside the Case Structures. An open box called Tunnel will be created on both main and upper Case Structure boundaries.

Step 19. Place the cursor on the output wire of F. You will find a hot spot. Click on it and draw two more wires to connect to A1 and A0, respectively. Thus we have connected F to A2, A1 and A0 to represent the binary number for the case 0. That is, the binary number for decimal number zero for [A2 A1 A0] is [0 0 0]. Now, if case 1 is selected, the binary representation of outputs A2 A1 A0 will be 0 0 1. Here, F should be connected to A2 and A1 for the 0s. For 1 we will consider the True (T) Constant by selecting Programming->Boolean-> True Constant option. That is, connect T to A0. For case 2, the binary output of [A2 A1 A0] is [0 1 0] and for case 3 the output is [0 1 1], and so on.

Step 20. Similarly, connect False (F) and True (T) constants to outputs (A2, A1, A0) for remaining cases to represent the corresponding binary number. We can follow the same technique for the inner lower Case Structure.

Step 21. Select another Select function and place in the main Case Structure. Connect output of 'Equal ?' correspond to input I7 to the 's' input of Select function. These are used to select the second-highest-priority request. Similarly, do it for the remaining inputs, except for I0.

Step 22. Connect 't' inputs of Select functions from the output of previous